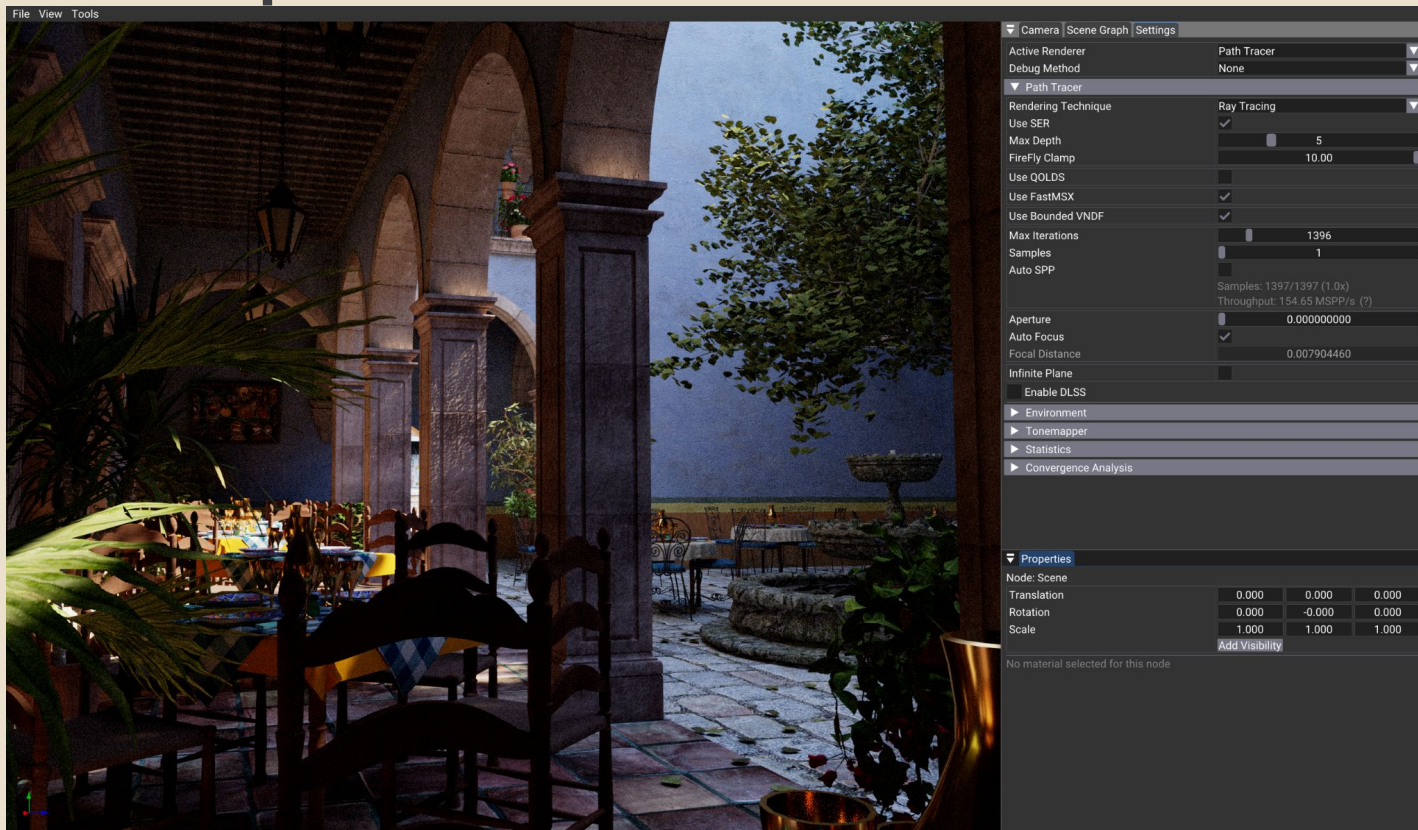


# MatForge - MS2

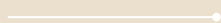
Cecilia Chen, Xiaonan Pan, Yiding Tian

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# Recap: what we do



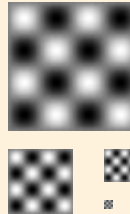
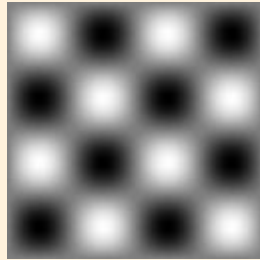
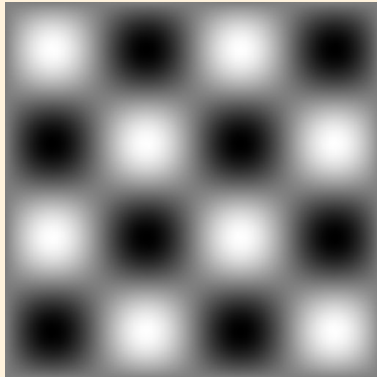
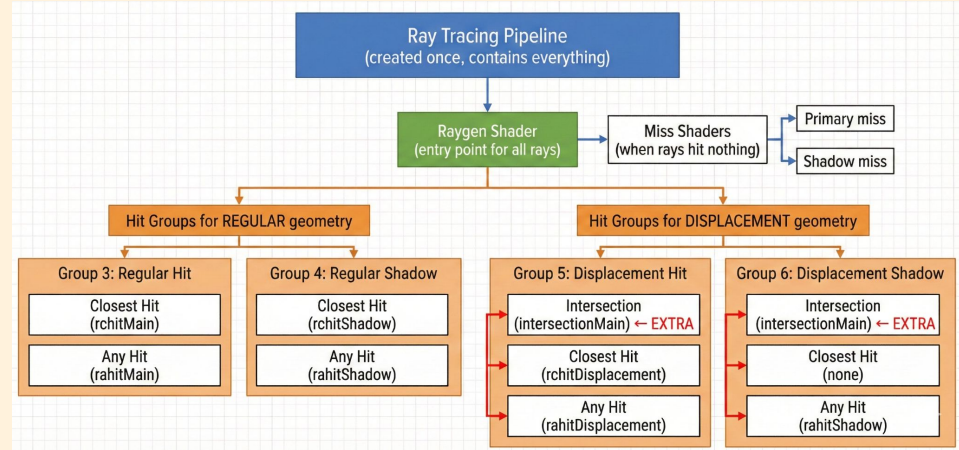
1



# RMIP - Cecilia

# Integrate RMIP displacement mapping into ray tracing pathtracer

- **Custom Intersection Shader**  
(`displacement_intersection.slang`)
- **PathTracer Pipeline Modifications:**  
Procedural hit groups for displaced geometry
- **RMIP Builder → PathTracer**  
connection established



Current displacement map test  
RMIP each layer in texture

## Next Milestone: Plan

- **BLAS Configuration**
- **Test Scene** - Verify RMIP builds and shader invocation
- **Visual Validation**- more complex scene with no visual effect



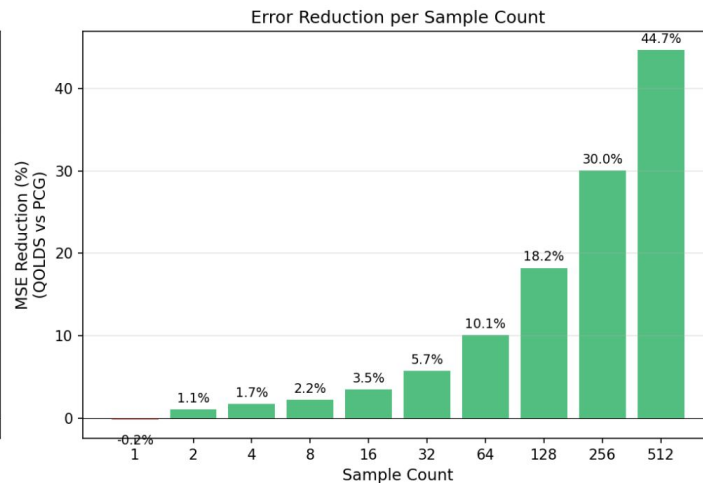
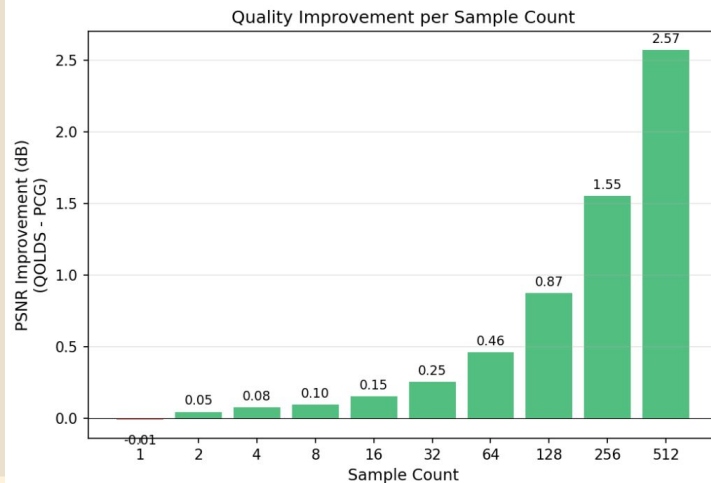
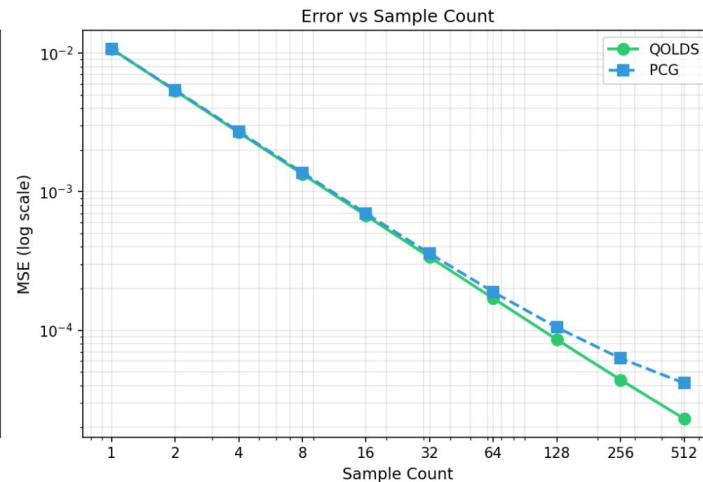
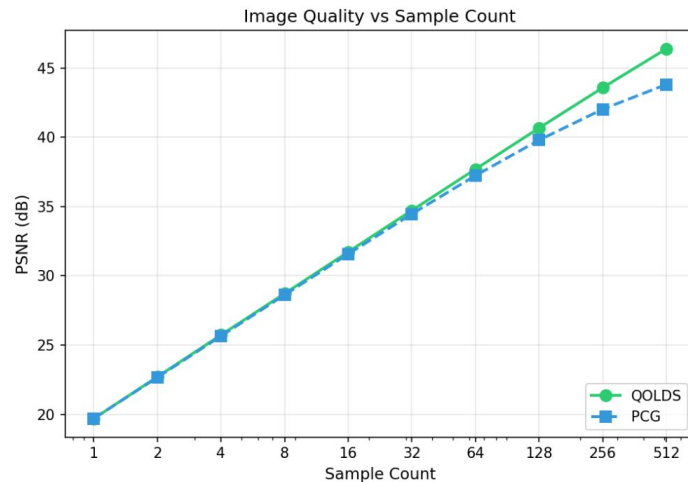


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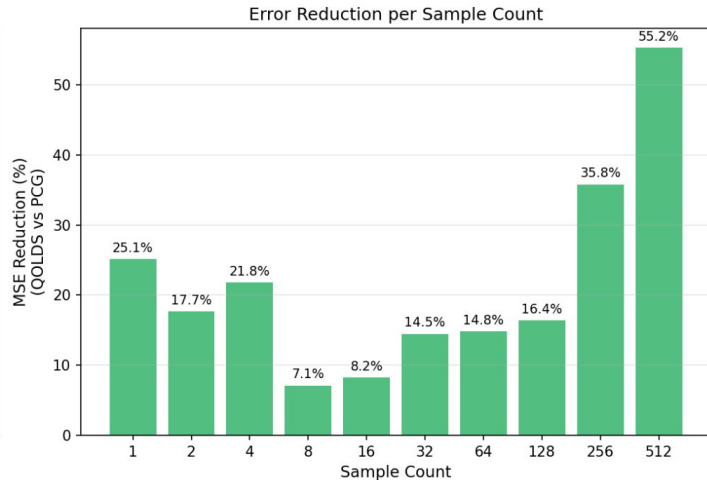
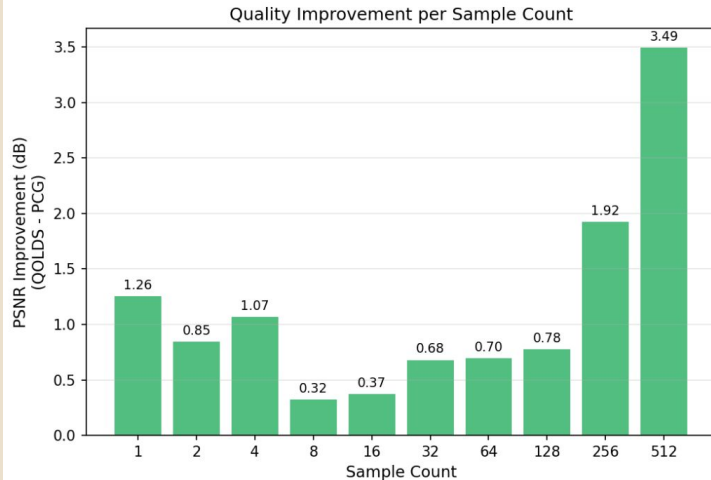
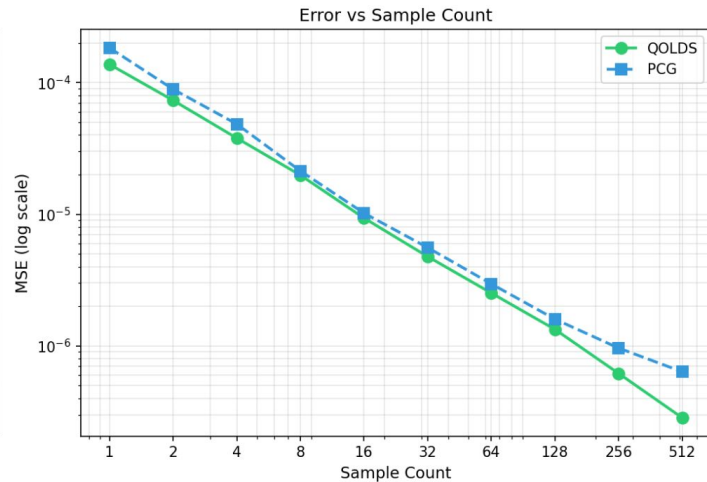
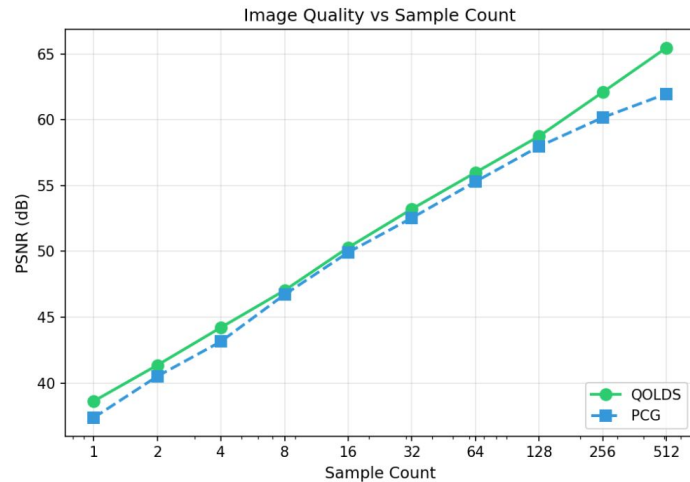
QOLDS - Yiding



## Convergence Analysis: QOLDS vs PCG

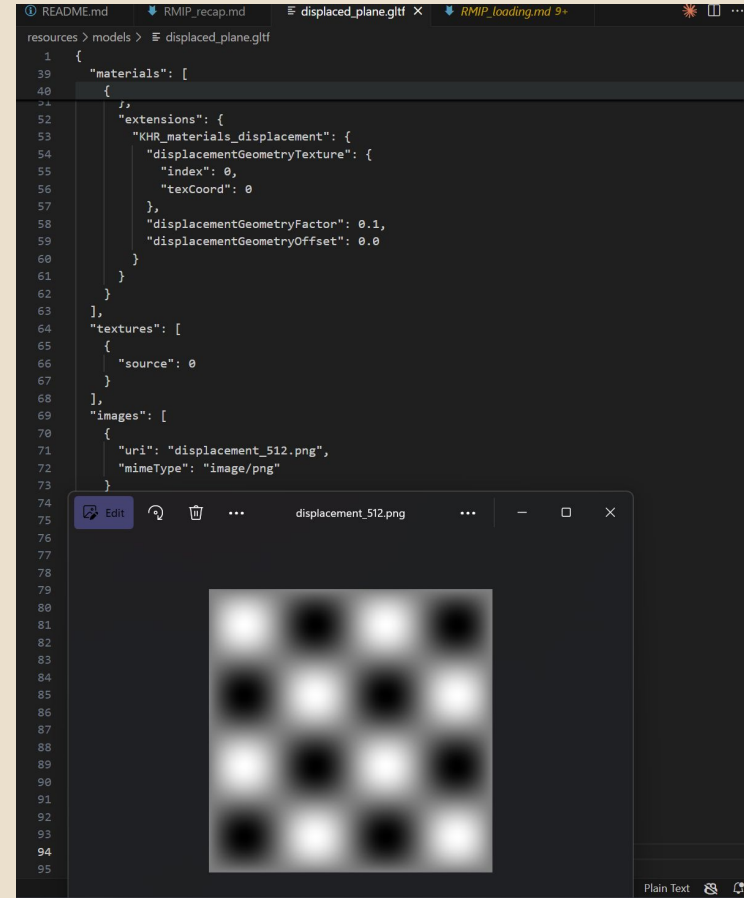


## Convergence Analysis: QOLDS vs PCG



# Other things I did

- Readme
- Project Website
- Multiple GLTF scene loading
- Transitioning to RMIP as QOLDS concludes





# Plan for Milestone 3

1

## Help with RMIP

Help with RMIP to find displacement maps and finish implementation

2

## Full integration

Full integration of the RMIP + QOLDS + FastMSX + Bounded VNDF pipeline

3

## Test Scenes

Create test scenes suitable for our pipeline

4

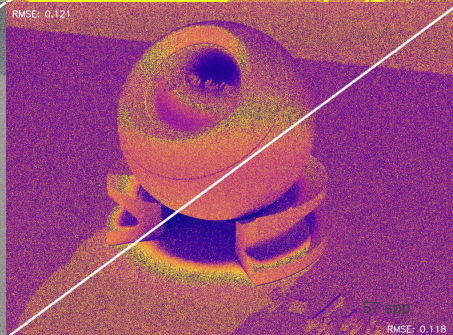
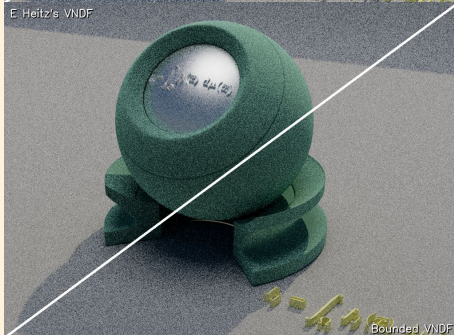
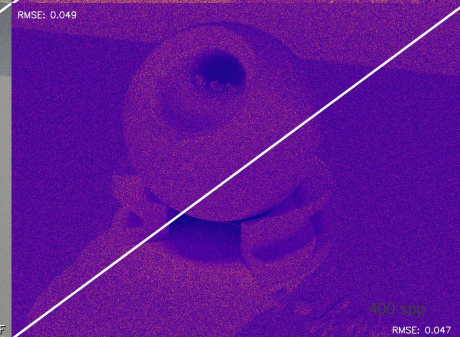
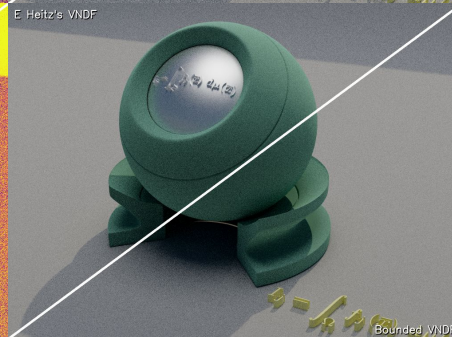
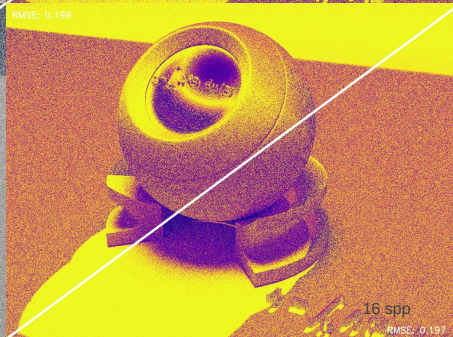
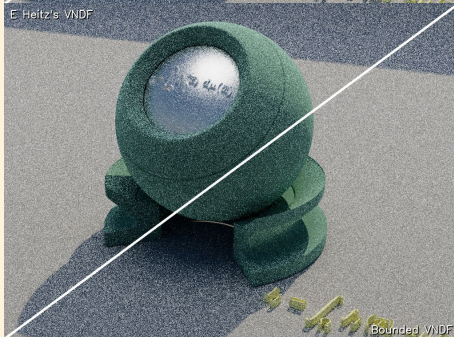
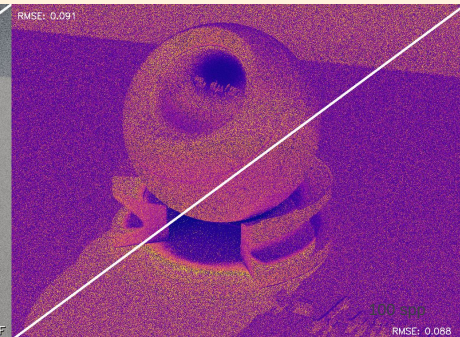
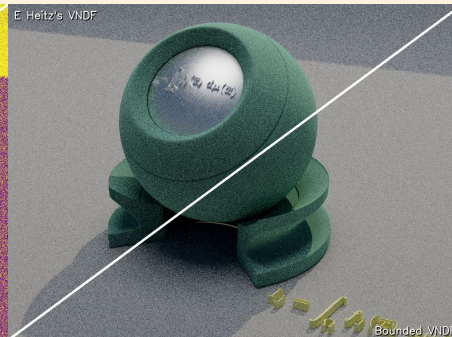
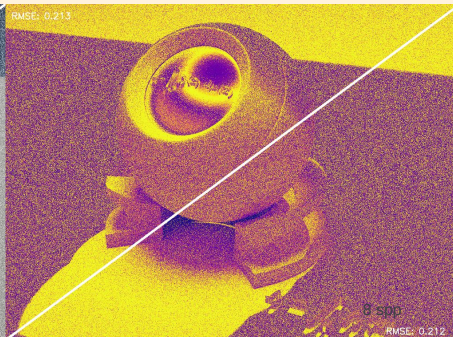
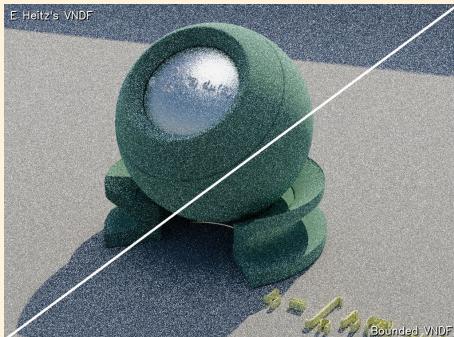
## Materials

Find materials that best showcases our capabilities

5

## Benchmarks

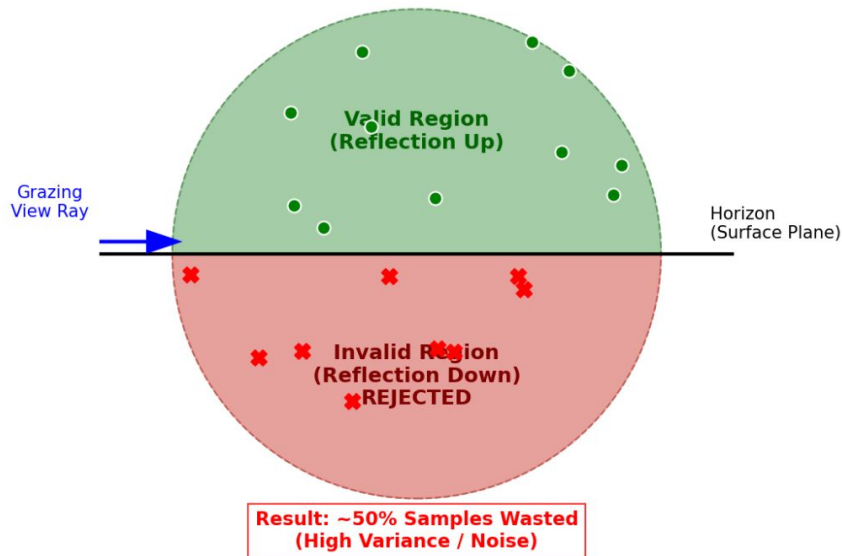
Benchmark and profile our implementation



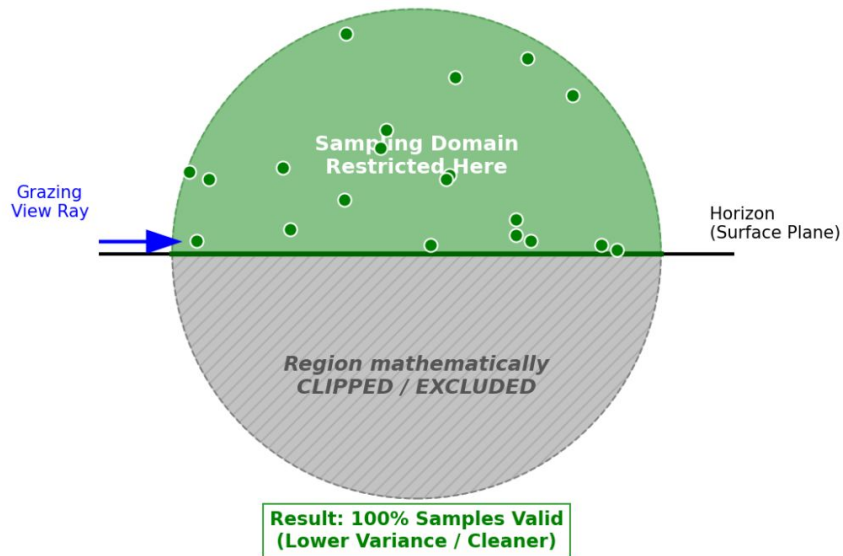
3

# Bounded VNDF Xiaonan

### Heitz's Standard VNDF (The Problem)



### Bounded VNDF (The Solution)



New Sampling Method + Corresponding PDF

# Plan for Milestone 3

- 1 Create the test scenes
- 2 Benchmark the performance with our features